Project Demo

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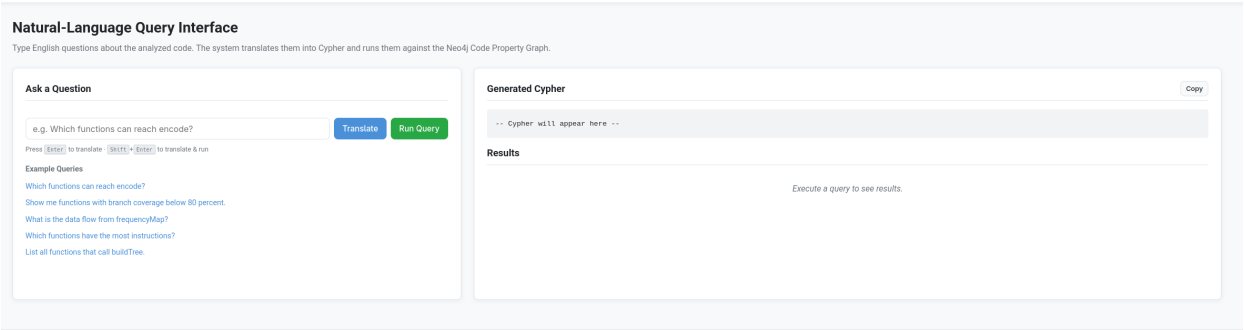
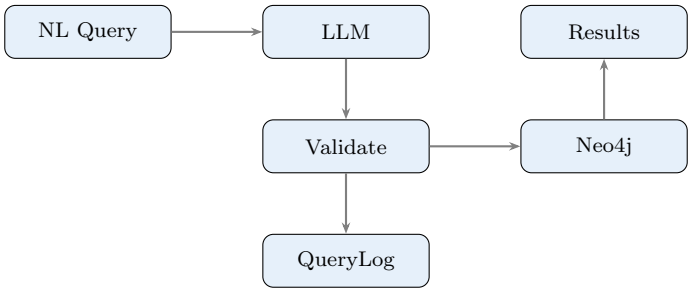
Queries

Call Graph: "Which functions can reach encode?" — Find transitive callers.
Coverage: "Show functions with branch coverage below 80%." — Identify under-tested code.
Data Flow: "What is the data flow from main:freq?" — Track inter-procedural dependencies.
Metrics: "Which functions have the most instructions?" — Locate hot spots.



Workflow



Project Dashboard

Static & Dynamic Analysis of an AI-Generated C++ Huffman File Compressor

Show system/stdlib functions

284	553	11	17
FUNCTIONS	BASIC BLOCKS	STORED QUERIES	VARIABLES

Top Functions by Instruction Count

FUNCTION	INSTRUCTIONS	LINE COV	BRANCH COV
`main`	110	100.0%	53%
`_Z9buildTreeSt3mapIciSt4lessIcESaISt4pairIKciEEE`	107	83.3%	72%
`_Z6encodeP4NodeNSt7__cxx1112basic_stringIcSt11char_traitsIcESaIcEEERSt3mapIcS6_St4lessIcESaISt4pairIKcS6_EEE`	59	71.4%	75%
`_Z7getFreqNSt7__cxx1112basic_stringIcSt11char_traitsIcESaIcEEE`	55	80.0%	90%
`_ZN4NodeC2EiPS_S0_`	17	25.0%	100%
`_ZN4NodeC2Eci`	15	100.0%	100%
`_ZN7ComparecIEP4NodeS1_`	12	33.3%	100%

Recent Queries

List all functions main calls.
OK 18716.78 ms

List all functions main calls.
OK 11721.99 ms

List all functions that call buildTree.
OK 16844.69 ms

Which functions have the most instructions?
OK 5249.1 ms

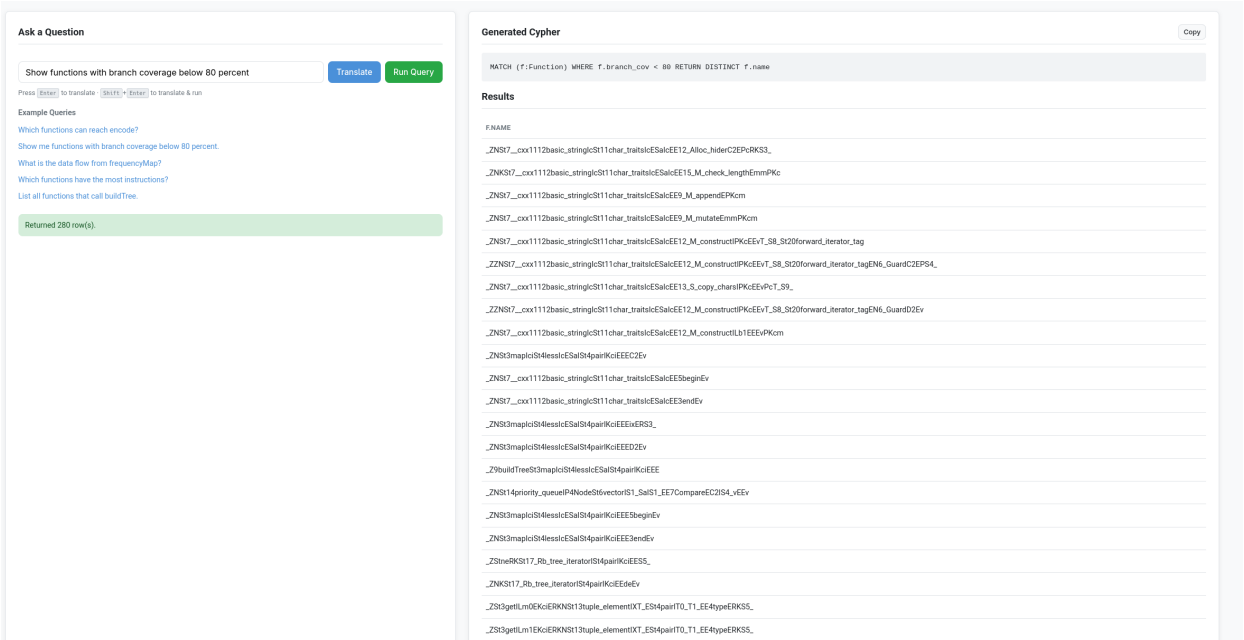
Which functions have the most instructions?
OK 5249.1 ms

Examples

Q1: "Show functions with branch coverage below 80 percent."

`MATCH (f:Function) WHERE f.branch_cov < 80 RETURN DISTINCT f.name`

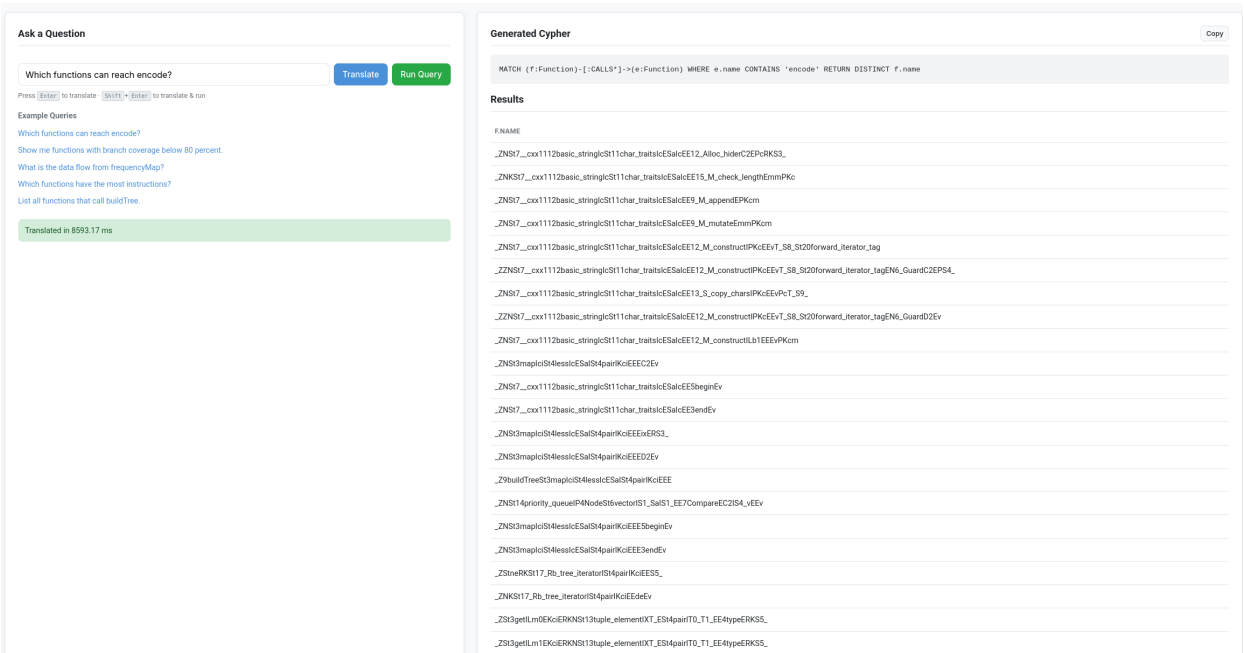
Result: 280 functions (mostly STL internals at 0%).



Q2: "Which functions can reach encode?"

`MATCH (f:Function)-[:CALLS*]->(g:Function {name: 'encode'}) RETURN DISTINCT f.name`

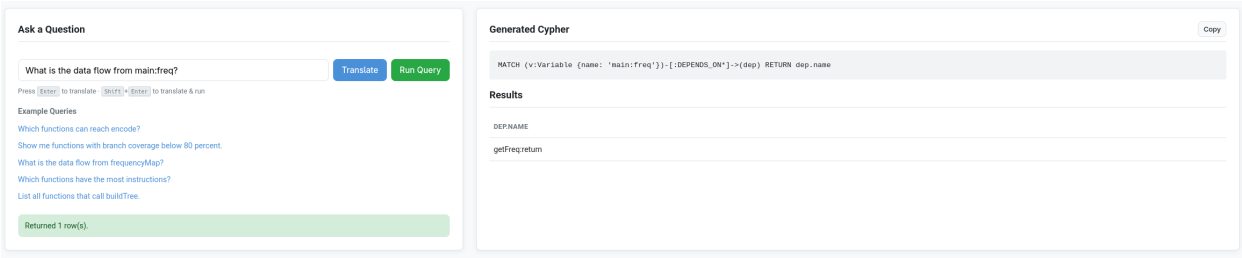
Result: `main` (via getFreq → buildTree → encode).

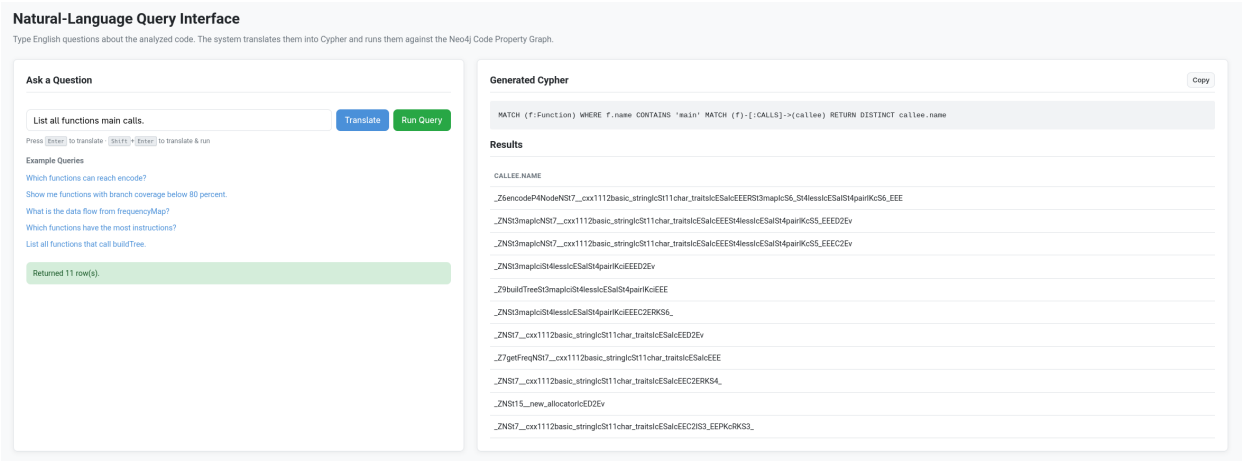


Q3: "What is the data flow from main:freq?"

`MATCH (v:Variable {name: 'main:freq'})-[:DEPENDS_ON]->(dep) RETURN dep.name`

Result: `getFreq:return`.



Extras

Auto-correction: Hallucinated names (e.g., "frequencyMap") fuzzy-matched to valid names (e.g., "main:freq").
Call Graph Viz: Interactive D3.js force-directed graph.
Function Explorer: Table of 284 functions with static + dynamic metrics; modal shows callers, callees, CFG.